

KARTHIK YANNABATHINA

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MERN Stack | JavaScript • TypeScript • Python | Open Source Contributor

SKILLS

Frontend	React, TypeScript, JavaScript, Redux Toolkit, Tailwind CSS, React Router, Vite, HTML5, CSS3
Backend	Node.js, Express.js, REST APIs, JWT, Socket.IO, Zod, Multer
Databases	MongoDB, Mongoose, PostgreSQL, Redis
AI / ML	RAG, BM25, Hugging Face Embeddings, Ollama, Local LLM Inference
Tools	Git, GitHub, Docker, Docker Compose, Postman, Jest, Swagger

PROJECTS

Grounded AI — Document Intelligence Platform | [Live Demo](#) • [GitHub](#)

React • Node.js • Express • TypeScript • MongoDB • Hugging Face • Ollama • Docker • RAG

- Built a **production-oriented RAG platform** that ingests PDF/Markdown documents and returns **grounded answers** using **local AI inference**.
- Implemented **hybrid retrieval (vector + BM25)**, **cross-encoder reranking**, local Hugging Face embeddings, and **Ollama LLM inference**.
- Fully **containerized with Docker**; designed for privacy-preserving, offline-capable document Q&A; in production environments.

TradeX — Real-Time Stock Market Simulation | [Live Demo](#) • [GitHub](#)

Node.js • Express • MongoDB • Redis • JWT • Google OAuth • Socket.IO • React

- Built a full-stack **real-time trading simulator** with live candlestick charts, portfolio tracking, and buy/sell execution.
- Implemented secure authentication (**JWT + Google Sign-In**), **Redis caching** for market data, and backend-heavy order matching logic.
- Designed **real-time updates** via Socket.IO and persistent portfolio state for an interactive trading experience.

Finance Tracker — Automated Expense Management | [GitHub](#)

MongoDB • Express • React • Node.js • JWT

- Full-stack MERN expense tracker that **automatically records expenditures** after successful transactions.
- Implemented **JWT-based authentication** and seamless post-transaction expense updates with clean state management.

OPEN SOURCE CONTRIBUTIONS

archive-movie-browser — Merged PR #100 | [Live Demo](#) • [Pull Request](#)

URL State Synchronization • Browser History API • Testing

- Fixed critical navigation bug where filters lived only in component state, breaking **shareable links** and the browser Back button.
- Implemented full **URL state synchronization** using pushState/replaceState/popstate with validation and safe fallbacks (+117 / -8 lines, 77 tests passing).
- Maintainer feedback: *"Shareable links and a working Back button are a big deal for an app like this."* — amponce

Contribute-To-This-Project (2.5K+ stars) — Merged PR #4767

- Added personal developer card following repository contribution guidelines; passed all automated checks and merged to master.

EDUCATION

Bachelor of Technology in Computer Science & Engineering

P P Savani University • 2023 – 2027

ACHIEVEMENTS

- Solved **100+ LeetCode** problems (Data Structures & Algorithms) | [LeetCode Profile](#)
- Merged open-source contributions with measurable impact (+117 lines, 77 tests) and positive maintainer feedback
- Built and deployed production-ready full-stack applications with live demos (RAG platform, real-time trading simulator)